

Human Factors Engineering, IE-1061

Project Lab 1: Product Description Fall 2025

Group Members: Arth, Alexa, Aleia, Amaris, Irving

Objective:

To gain experience in breaking down a product into user elements

Overview:

You and your team will select an object, tool, product of some kind to design or redesign. You can redesign one product, combine multiple products, or design a completely new product. Before jumping in and designing the product, you must conduct a detailed product and user description along with a task analysis and flow process chart for how the product was or should be manufactured. These latter two will be accomplished in project lab 2.

From the Memo:

<u>Memo</u>	<u>Date</u>	<u>Description</u>	<u>Date Due</u>
ProjLab1 – Detailed Project Description; Manufacturing process analysis and flow chart – final is not required yet...draft only so far - get as far as you can this week. Pick groups (choice within lab – 3-5 is best)	4/5 Sept. 2025	- Provide a complete description of the product, its users and the environment in which it is used. This is Stage 1 of the system design process - Take pictures or sketch (or some other rendering) the product for use in the description. - Develop a <u>Flow Process Chart</u> for how you believe this product is currently manufactured (IE-1054 and IE 1052 may be your reference point here) - Capture anything you learn from a marketing perspective.	18/19 Sept. 2025

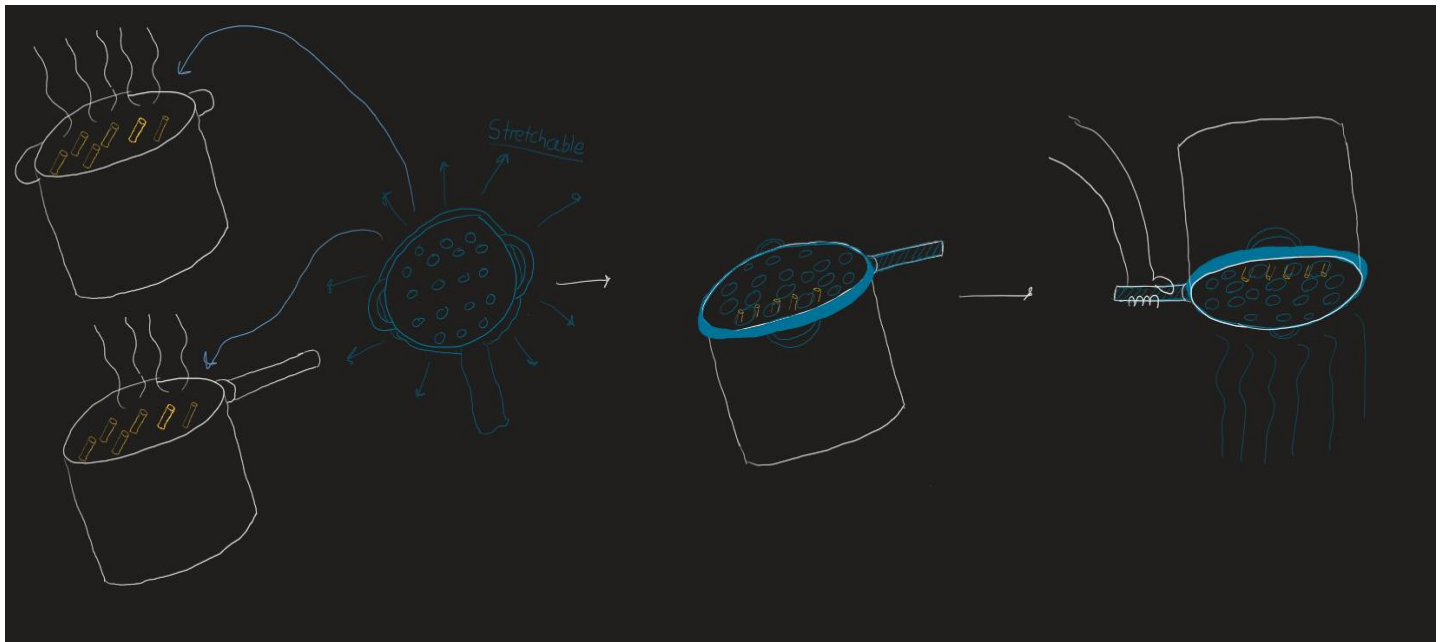
Instructions/Deliverables:

- Take several photographs of your product (or some other type of rendering if you are designing a new product), both alone and while in use. Save your pictures! You will use these pictures in your report to describe the current product.

Below we have photos of the product we are choosing to make an innovation for. The images show a product that is able to clip onto the pan, catch pasta, and strain water. However, this doesn't make the process as easy as we'd want it to be. Our product innovates upon the original idea in a way that makes all steps of the "draining boiling pasta water" easier. The goal is to have silicon stretching material that is able to fit the sizes of small to even bigger pots. It should fasten around the entire size of the pot securely to allow you to be able to flip the pot at any angle to get the water out while reducing the amount of water that may splash you. Further our design includes two different types of handle grip fasteners. One for pots that have two handles and the other for pots that have one long handle (more on pans). For pots that have handles that also heat up to a high temperature that doesn't allow you to touch safely the silicon handles from our product will easily be able to wrap around and secure itself around the handles allowing you to pick up the hot pot without it burning you.

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2. Determine the general objectives of your product and its performance requirements (do not forget HFE needs). What particular features does your particular product have over similar products? Develop a Market Requirements Document (MRD). An example is on this website: ([Market Requirements Document \(MRD\) | Definition and Overview \(productplan.com\)](#))

Objective :

The primary objective of our product is to simplify and enhance the cooking process, particularly for food items that require boiling and draining. The product must be safe and efficient for everyday use. It should eliminate the need for multiple strainers, while minimizing the risk of burns or spills.

Requirements:

- The product must withstand exposure to boiling liquid
 - Product must be simple to attach, detach, or position on different cookware without requiring excessive force.
- Differentiating Feature:
- Adaptable to Different Pots and Pans

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- Suitable for different food Sizes
Market Requirement Document

Requirement	Detail
Target Market	Kitchenware and Cooking accessories industries focused on making utensils for home use, collage students and small-scale kitchen environment. We believe it it worth pursuing because it ensures the safety and comfortability of users as well it condenses the need for several kitchen utensils into one, making it more practical.
Potential Revenue	The ARPU of the kitchenware company was valued at aveargely 69.04 billion dollars and has about 72.8 billion potential customers. Which establishes a massive TAM value. If our product captures even about 0.01% of this TAM it would represent a potential revenue of about \$6.9 million.
Buyers and User Persona	Primary Users - College students who have smaller living spaces and need more condensed utensils which are more efficient - Families that cook a lot and prefer more time efficient utensils. Secondary buyers - Kitchenware retail stores like supermarkets - Online kitchenware sites
Market Needs	- Reduce clutter in kitchens by eliminating multiple strainers. - Provide a safer alternative to traditional strainers, especially for novice cooks or elderly users. - Deliver a cost-effective product that combines functionality, safety, and convenience.

Amaris

Irving

Discussion/Issues to Think About:

As you complete the first stage of your product human factors engineering/ergonomics-based design or redesign, discuss some of the issues/things your team will take into consideration in this design or redesign. Had you thought of these things prior to conducting this lab? Issues such as target customer base, ease of manufacturing, potential cost to manufacture, product liability, durability, ease of use, maintenance, etc.

Note to project teams:

While I ask for complete and detailed descriptions, it is understood that in this first project lab, everything is still in draft and is very changeable.

- Determine who your user(s) will or should be. Make considerations for the following: young, old (give ages), handicapped, left hand vs. right hand, expert vs. novice, etc. and define them.

3.Users/Stakeholders

Age Range	Characteristics	Pain Points/Needs	Design Considerations
Young (under 30)	Kitchen novices, occasional cooks	Easy and intuitive use, quick operation	Simple clip-on, light weight, clear handle

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			orientation(reduce confusion even without manuals or instructions)
Mid-aged (30-55)	Regular home cooks, some experience, efficiency-oriented	Durable, stable, handles larger portions	Easy to clean and organize (stainless steel/heat-resistant plastic?), secure attachment to different size of pot/bowl
Elder (above 55)	Reduced hand strength and reactivity, possible arthritis	Comfortable grip, light weight, easy to maneuver	Ergonomic, anti-slip handle, less force needed to hold and pour

Different User Groups	Characteristics	Pain Points/Needs	Design Considerations
With hand disabilities	Limited grip strength or balance	Minimal effort required to hold stable when pouring. Better can operate with one hand/do not require fully both hands	Thick, contoured handle, single-hand operation, anti-slip/static operation
Left-handed	Left-hand dominant	Handle and strainer usable with either hand	Symmetrical handle design or reversible orientation
Novice cooks	Inexperience with kitchen tools	Clear, safe, predictable, easy to use or direct/concise instructions	Visual cues for placement, easy clip-on mechanism, self-stability
Expert cooks	High kitchen skills, possible more cooking requirements	Large capacity, frequency of use.	Durable, heat-resistant materials and appropriate force distribution design for longer lifetime, reduce wearing and damage

- Develop a Manufacturing Flow Process Chart given your knowledge of IE 1054 (Productivity Analysis) and IE 1052 (Manufacturing Systems). Please arrange to meet with your TA or with Dr. Kazemi to discuss your product's process chart if needed, (it is recommended that you do this to make sure you are on the right track with your ideas). Towards the end of the term, you will be likely to revise the manufacturing process chart as your idea develops and given the changes that you will make. Make sure that you specify the types of operations that you should use to manufacture your product!! Keep in mind this process chart is on the original product or products (if you are doing a redesign). If you are doing an initial design, you will have to develop this from scratch with a good bit of homework around what you think would be needed to manufacture your product.

Manufacturing essentially involves using injection molding to create the main silicon component, then using injection molding to create the plastic clips:

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